Lecture 1. Introduction

Survival Data, Censoring, and Truncation

1 Course Overview

- Textbook: Klein & Moeschberger, *Survival Analysis: Techniques for Censored and Truncated Data*, 2nd ed. (cited below as K&M). These notes follow its Chapters 2–5, 7–9, 11 and 12.
- The emphasis is on methodology and application, with part of the underlying theory (why the methods work and how they were developed).
- Prerequisites: basic probability theory, statistical inference, and R programming.

2 What Is Survival Data?

Definition 1.1: Survival data

Survival data track the *time until an event of interest occurs*:

$$\text{Exposure (clock starts)} \xrightarrow{\text{survival time}} \text{Event (clock ends)}.$$

Examples: time from diagnosis to death; time from treatment to cancer recurrence; time to failure of a machine or component; time to staff promotion; time to divorce; time to unsubscription.

Example 1.2: Worcester Heart Attack Study (WHAS100)

Survival times following hospitalisation for a heart attack (MI). A snapshot:

ID	Female	Age (years)	Follow-up (years)	Event (death)
16	0	39	7.2	0
66	1	91	1.0	1
71	0	65	5.1	0
55	0	80	5.3	0
61	0	90	0.3	1

Question: what is the mean survival time after hospitalisation?

- *Wrong approach 1* (use only subjects with known survival times): $(1.0 + 0.3)/2 = 0.65$. This is an **underestimate**, because we ignored the people who survived longer.
- *Wrong approach 2* (pretend follow-up time = survival time): $(7.2 + 1.0 + 5.1 + 5.3 + 0.3)/5 = 3.78$. Also an **underestimate**, because IDs 16, 71, 55 actually survived *longer* than their follow-up times.

Acknowledging censoring. Writing censored times with a "+": 7.2^+, 1.0, 5.1^+, 5.3^+, 0.3. For ID 66 we know the patient survived exactly 1 year after MI; for ID 16 we only know that the patient survived *at least* 7.2 years. The data carry **incomplete (partial) information**.

Remark 1.3: Why both naive estimates are biased downward

Let T_i be the true survival time and C_i the follow-up (censoring) time, so we observe $Y_i = \min(T_i, C_i)$. Since $Y_i \leq T_i$ always, $\mathrm{E}(\bar{Y}) \leq \mathrm{E}(T)$ (approach 2). Approach 1 keeps only $\{T_i \leq C_i\}$, i.e. it conditions on the event time being *small enough* to be seen, which preferentially discards long survivors.

3 What Is Special About Survival Data?

- Classical mathematical statistics is built on i.i.d. data, e.g. n i.i.d. copies $T_1, \ldots, T_n$ of the survival time.
- The **incompleteness** of survival data distinguishes survival analysis from traditional methods.
- Formulating a survival problem requires three ingredients:
  (1) the **event of interest**;
  (2) the **time origin**;
  (3) the **time scale**.
- Incompleteness is realized as either **censoring** or **truncation**. Common types: right censoring (RC), left censoring (LC), interval censoring (IC), left truncation (LT), right truncation (RT).

4 Types of Censoring

4.1 Right censoring

Definition 1.4: Right censoring (RC)

For a right-censored observation, the exact event time is unknown; all we know is that the event happens *after* a certain time point: $T_i > C_i$.

Example 1.5: Time to death in a breast cancer trial

T_i = time to death from randomization (receipt of treatment); the time of randomization is the time origin. If the study ends 9 months after the patient's treatment date and the patient is still alive, T_i is right censored at 9 months: $T_i > 9$.

4.2 Left censoring

Definition 1.6: Left censoring (LC)

For a left-censored observation, all we know is that the event happened *prior to* a certain time point: $T_i < C_i$.

Example 1.7: Time to first use of marijuana

Question: "When did you first use marijuana?" Possible answers:

- the exact age (uncensored);

- "never used it" (right censored at the current age);
- "used it but cannot recall when" (left censored at the current age).

When both left and right censoring are present in the data we speak of **double censoring**.

4.3 Interval censoring

Definition 1.8: Interval censoring (IC)

For an interval-censored observation, all we know is that the event happened within an interval: $T_i \in (L_i, R_i]$.

Example 1.9: Time to tumour recurrence

After breast cancer surgery, MRI is performed every 5 months. If the tumour is absent at month 10 and detected at month 15, then $10 < T_i \leq 15$.

Remark 1.10

RC and LC are special cases of IC: right censoring at C is $T \in (C, \infty)$, left censoring at C is $T \in [0, C)$.

5 Types of Truncation

5.1 Left truncation

Definition 1.11: Left truncation (LT)

Only individuals who "survive" a sufficient time are **included** in the sample. If V_i is the entry (truncation) time, T_i is observed only if $T_i \geq V_i$, i.e. we observe the conditional distribution of T_i given $T_i \geq V_i$.

Example 1.12: Longevity of elderly residents of a retirement community

An individual must survive to a sufficient age to enter the retirement center; anyone who died earlier never enters. The data are left truncated. Ignoring LT leads to **length-biased sampling**: long lifetimes are over-represented.

5.2 Right truncation

Definition 1.13: Right truncation (RT)

Only individuals who have *experienced the event* by a specified time are included in the sample: T_i is observed only if $T_i < R_i$.

Example 1.14: Age at onset of diabetes

Diabetic patients are enrolled and asked retrospectively for their age of onset. With R_i = age at the time of the study and T_i = age of onset, T_i is observed conditionally on $T_i < R_i$: a subject must already have diabetes to be included.

Example 1.15: HIV incubation time (Lagakos et al., 1988)

Data on 258 adults and 37 children infected with HIV who developed AIDS by June 30, 1986. With X = age at HIV infection and R = age at developing AIDS, the incubation time $R - X$ is known only for those who were infected *and* developed AIDS within the study window. This is right truncation.

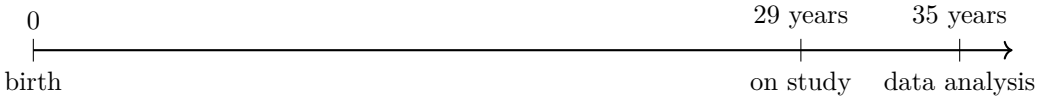
5.3 Censoring vs. truncation

Censoring	The event/subject *is detected*, but its value is not known completely: we only know it is less than and/or more than some number.
Truncation	A subject is detected *only if* its value is greater (or less) than some number; when detected, its value is completely known.

The key difference: a censored object is detectable, whereas a truncated object is not even detectable (it never enters the data set).

6 CVD Example: Time Scales, Truncation and Censoring Together

An observational study enrols subjects *without* heart disease to identify risk factors for cardiovascular disease (CVD); the event is CVD. Consider a subject who enters at age 29; data are analyzed when the subject would be 35.



Two time scales are possible depending on the origin:

- origin = birth: T_i^1 = age at CVD;
- origin = study entry: T_i^2 = years on study until CVD.

Both may be right censored (subject still CVD-free at the end of study). The first scale T_i^1 is usually more interesting.

Scenario	T_i^1 (age)	T_i^2 (time on study)
CVD at age 34	$T_i^1 = 34$ observed *given* $T_i^1 \geq 29$: **LT at 29**	$T_i^2 = 34 - 29 = 5$: no truncation, no censoring
No CVD by 35	$T_i^1 > 35$ given $T_i^1 \geq 29$: **LT at 29 and RC at 35**	$T_i^2 > 6$: **RC**, not LT
Enrolled at 30, already had CVD before 30	$T_i^1 < 30$: **LC at 30**	—

7 Definitions and Censoring Patterns

- **Failure time** (survival time, event time): the length of "time" to an event.
- **Time origin**: objectively defined and scientifically meaningful; ideally all subjects are comparable at the origin.
- **Time scale**: usually clock/calendar time (seconds, hours, weeks, months, years), but others are possible, e.g. QALY (quality-adjusted life years).

Definition 1.16: Equal or unequal censoring times

- **Type I censoring**: all subjects enter at the same real time and all censoring times are equal.
- **Generalized Type I censoring**: with staggered entry the censoring times differ, but each is known at entry.
- **Random censoring**: losses to follow-up (death from other causes, relocation, withdrawal of consent); censoring times are known only for those actually censored, and they differ.

It is crucial that the actual censoring *times* are recorded, not just the fact that a subject is censored.

8 Possible Simple (Naive) Methods

1. **Dichotomise at a pre-specified t_0.** Survivors to t_0 are "successes", deaths before t_0 "failures"; use binary-data methods (chi-square test, logistic regression).
   Disadvantages: difficulty with censoring before t_0; loss of efficiency from concentrating on a single point of the survival curve; arbitrary choice of t_0.
2. **Take logs and use the linear model.** Disadvantages: possible undue dependence on short failure times; the log-normal distribution may not be appropriate; still cannot handle censoring.

Both can be useful as a prelude to more complex analyses.

9 Summary

- Six types of incompleteness: RC, LC, IC, LT, RT (plus double censoring). LT and RC are the most common sampling schemes.
- **Censoring only**: T_i may not be observed directly; it is known to be $>$ or $<$ some value, or to lie in an interval.
- **Truncation only**: T_i is observed conditionally on $T_i >$ or $<$ some value.
- Mixtures of censoring and truncation are more complex.